Aryuemaan Kumar Chowdhury

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.08
Sri Chaitanya Junior College	Class XII (TSBIE)	2020	67.30%
SMP Universal School	Class X (CBSE)	2018	42.00%

PROFESSIONAL EXPERIENCE

Founder & Lead AI Engineer | OSCOWL AI (Jan 2024-Present)
- Developed **QuantaFlow** on Azure, cutting quant strategy iteration cycles by 70% for 500+ developers.
- Engineered a real-time market-microstructure LOB dashboard and built a **DST-funded RL flight controller**.

Research Assistant | TOC0DE Lab, IIT Hyderabad (Sep 2025-Present)
- Built Python/MATLAB AI topology suites (HoneyTop90), cutting structural mass by 35% using GPU solvers.
- Fused **CUDA kernels** across 5+ ML architectures, reducing inference latency by 28% for rocket components.

Autonomous Systems Engineer | Suzuki R&D / TiHAN, IITH (May 2023-Jun 2024)
- Developed **YOLO + LiDAR-SLAM** pipelines on Jetson Orin, boosting collision accuracy by 55% in CARLA.
- Optimized GPU scheduling to sustain 30 Hz real-time mapping, validated on the KITTI dataset.

SKILLS

ML / AI: Large Language Models (LLMs), Small Language Models (SLMs), Cross-Modal Tensor Fusion, Transformer Architectures, Speech Synthesis (Real-Time Voice Preservation), Reinforcement Learning (RL), Natural Language Processing (NLP), Deep Learning Frameworks, RAG Architecture, MLOps Pipelines
Languages Data Eng: Python (Advanced/PyPI Package Publisher), PyTorch, C++ (STL), CUDA (Parallel GPU Processing), MATLAB, High-Throughput Audio Streams, Real-Time Audio Encoding/Decoding, PyPI Package Deployment, Git/Version Control
Cloud/DevOps: Model Optimization, Edge AI Deployment, TensorRT, Model Quantization, Cloud Compute Clustering, Enterprise Agent MLOps, Custom Model Training Frameworks, Linux Core, Continuous Integration/Continuous Deployment (CI/CD)
Engineering/Physics: Aerothermodynamics, Method of Characteristics (MoC), Computational Fluid Dynamics (CFD), Rocket Nozzle Flow-Field Generation, Bio-Mimetic Flight Dynamics, Autonomous Control Systems, CAD File Automation (DXF/JSON Export)

RELEVANT COURSES

Machine Learning & Applications, Material Data Science, Topology Optimization, Additive Manufacturing, Automotive Materials

PROJECTS

- **Vocalis Speech AI:** Created a real-time, voice-preserving speech translation system that retains vocal profiles.
- **AeroProp-X:** Published a PyPI Python package using Method of Characteristics for CAD-ready rocket nozzle contours.
- **Bio-Mimetic Flight Control:** Built ML algorithms based on biological patterns for autonomous flight in turbulence.
- **NumeraPi an AI Studio:** Built fully integrated workspace, which generate code snippets, with provision the underlying Kubernetes architecture, parse the dependencies, and deploy the environment.

AWARDS AND RECOGNITIONS

- **Best Startup Award:** Computer Society of India, Hyderabad Chapter (Jul 2024)
- **Entrepreneurial Excellence Award:** KL University Hyderabad, for OSCOWL & Co (Apr 2024)
- **World Rank 201:** IEEEXtreme Competitive Programming Hackathon (Sep 2022)
- **Startup Pitch Winner:** KL University CIIE, Start-up Rendezvous'21 & Innovation Day (Oct 2021)

RESEARCH PUBLICATIONS

- **Predicting Steel Fatigue Life from Micrographs Using Physics-Informed Deep Learning**
  arXiv preprint arXiv:2607.28695, 2026
- **Wiola 13M, a Gated Spiral Attention Architecture for Parameter Efficient Small Language Models**
  arXiv preprint arXiv:2608.14604, 2026
- **The Wiola Architecture for Efficient Small Language Models**
  arXiv preprint arXiv:2607.01394, 2026
- **Dual-Lane Voice-Preserving Real-Time Speech Translation: A System Architecture for Cross-Lingual Speaker Identity Retention**
  International Conference on Artificial Intelligence in Information and Communication Technology, 2026
- **Cross-Modal Attention and Tensor Fusion for Deep Audio-Visual Emotion Detection**
  ICSCST, vol. 23, p. 5, 2026
- **Theoretical Framework for AI-Powered Signal Compression in Deep Space Communication: Comprehensive Mathematical Analysis and Performance Bounds**
  IEEESpace, p. 5, 2026
- **Physics-Informed Generative Design for Nonlinear Topology Optimization of Aerospace Bracket Assemblies Compliant with ASTM Standards**
  Engineering Archive, 2026